CSE423 / COMPUTER GRAPHICS PROJECT

RAKKHOSH MUST DIE

A 3D tower-defense game inspired by *Orcs Must Die*

THE PREMISE

The player defends a central rift against seven escalating waves of attacking monsters. Movement, ranged combat, traps, automatic turrets, and collectible abilities must be combined to stop enemies before they reach the core. The final encounter is a 20-HP Master Boss supported by harder enemies.

07 enemy waves	**03** weapons
03 powerup types	**100** initial core HP

01 / PLAYER EXPERIENCE

Defend, reposition, repeat. Enemies enter from randomized arena corners and follow defined pathways toward the rift. The player can intercept them directly or slow their advance at chokepoints.

1. **ENGAGE** — Use pistol, crossbow or shotgun
2. **FORTIFY** — Place slowing or damaging traps
3. **ADAPT** — Collect timed abilities and reposition
4. **SURVIVE** — Protect player and core; reach Wave 7

02 / THREATS AND DEFENSES

Enemy	Profile
Easy (green)	1 HP, speed 7.0
Medium (orange)	2 HP, speed 6.0
Hard (red)	4 HP, speed 5.0
Master Boss	20 HP, speed 4.0

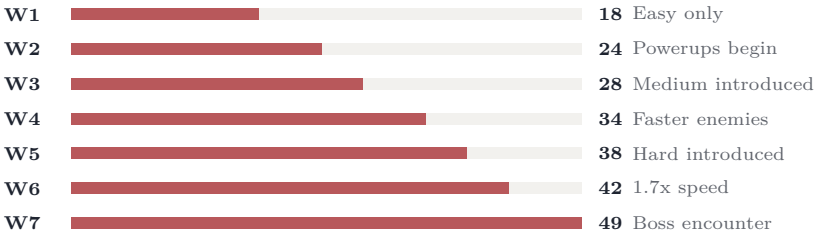
Three weapon choices

Pistol: rapid basic shots. Crossbow: powerful piercing shots. Shotgun: two spread pellets.

Layered defenses

Slowing traps, animated damage-dealing spikes, and wave-upgraded auto-targeting turrets.

03 / ESCALATING WAVES



Wave 7 includes 45 regular enemies, one Master Boss, and three additional Hard enemies. Its Hard enemies also fire projectiles.

Inside the Game Systems

How rendering, real-time simulation and player decisions work together.

RENDERING

OpenGL + GLUT

Spheres, cylinders, cubes and polygons form the 3D characters, arena, weapons and effects. Orthographic projection adds the on-screen HUD.

SIMULATION

State-driven game loop

`animate()` updates movement, gravity, enemies, bullets, collisions, active effects and wave progression.

ENEMY AI

Waypoints + aggro

Enemies move along predetermined path segments, may prioritize a nearby player, and damage the rift when they arrive.

COMBAT AND INTERACTION

Projectile motion is updated by velocity vectors, while hit tests compare projectile and enemy positions. Turrets locate nearby targets and rotate to fire at them; traps apply proximity-based slowing or damage. Player motion uses facing-angle trigonometry and gravity-driven jumps.

Timed powerups

Speed Boost	2x movement for 15 s (**1**)
Super Jump	2.5x jump for 15 s (**2**)
Roundhouse	360-degree firing for 7 s (**3**)

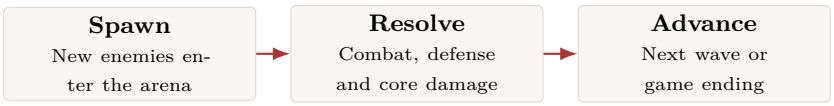
PLAYER CONTROLS

Action	Input
Move / turn	W/S, A/D or arrow keys
Jump / fire	Space / X
Weapons	4 Pistol; 5 Crossbow; 6 Shotgun
Deploy trap	T
Camera / menu	C / M
After game over	R to main menu
Exit gameplay	Esc

Readable status display.

The HUD provides player health, wave, enemy count, score, weapon selection, trap inventory and remaining powerup time.

HOW A ROUND IS RESOLVED



VICTORY

Defeat the 20-HP Master Boss in Wave 7. The game then clears remaining enemies and displays the final score and wave information.

DEFEAT

Game over occurs if player HP falls to zero or the core's HP reaches zero. Each enemy reaching the core inflicts 10 damage.

Implementation and demonstration. The game uses Python 3 and PyOpenGL with GLUT/freeglut. A gameplay demonstration should check weapon switching, turret and trap effects, powerup timers, wave progression, boss victory and both loss conditions. The README documents the intended behavior; no measured performance or playtest statistics are supplied.

Conclusion. *Rakkhosh Must Die* reinterprets the action and defense mechanics of *Orcs Must Die* in a custom-built 3D arena. Its combination of real-time graphics, collision logic, automated defenses and progressive enemy behavior creates a compact but complete game-design project.

Run: `pip install PyOpenGL PyOpenGL_accelerate` → `python orcs.py`